



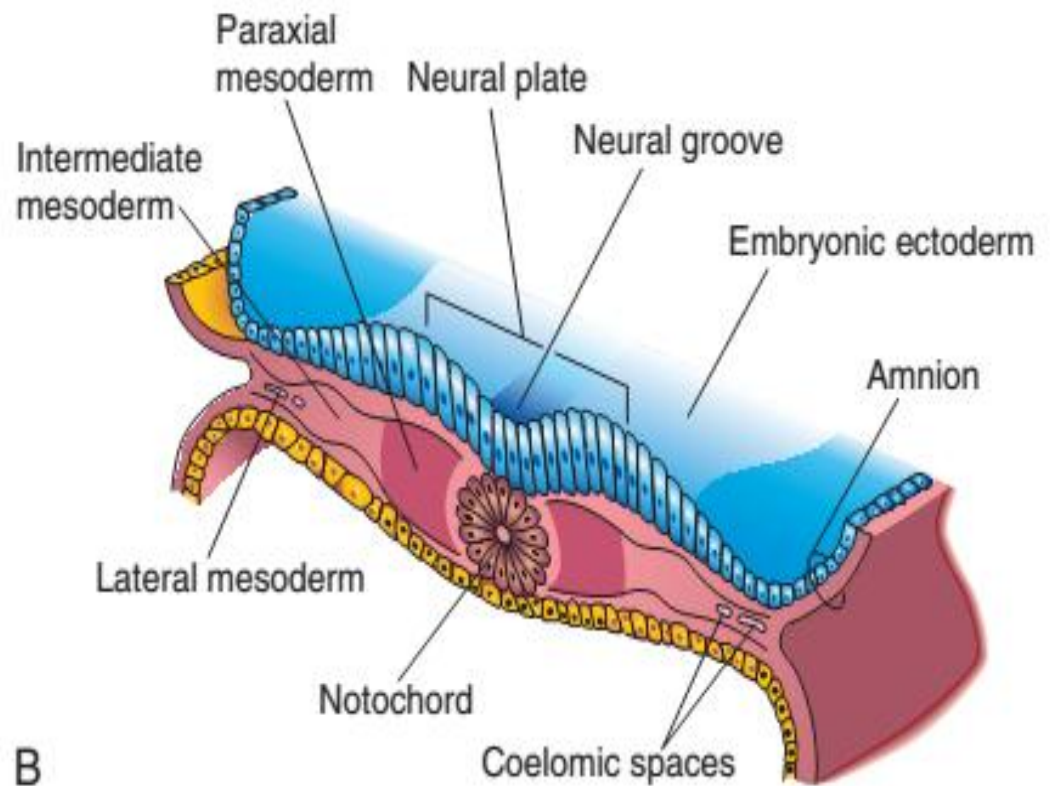
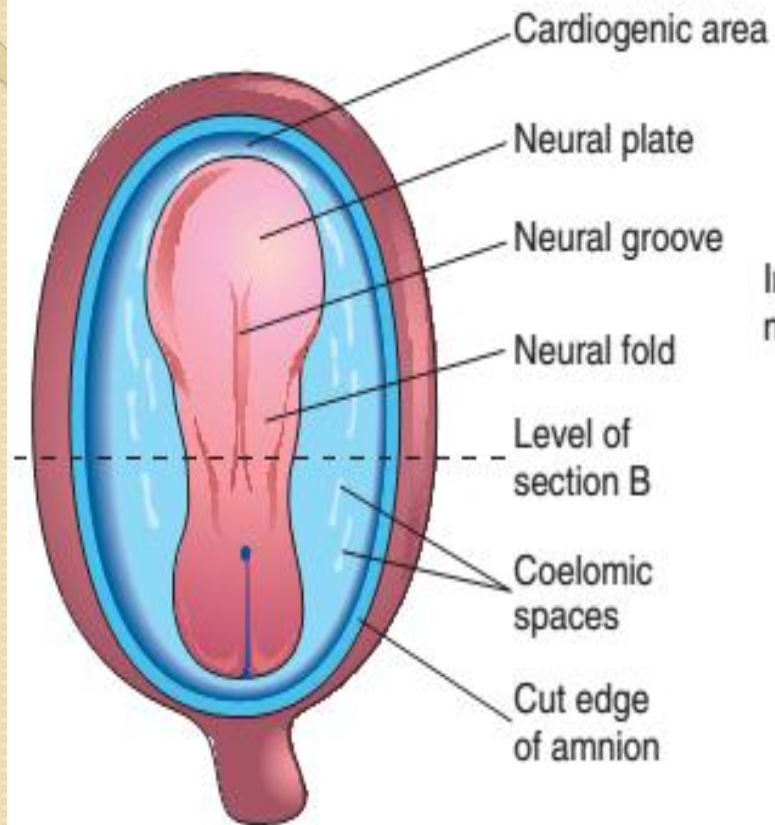
Development of the Skeletal System

SOLOMON T.

Skeletal System

- It develops from **mesodermal** and **neural crest cells**
- As the notochord and neural tube forms
- Embryonic mesoderm on each side of them proliferates
- **Form a thick longitudinal columns** of **paraxial mesoderm**
- Each column is **continuous with intermediate mesoderm**

Development of Skeletal system



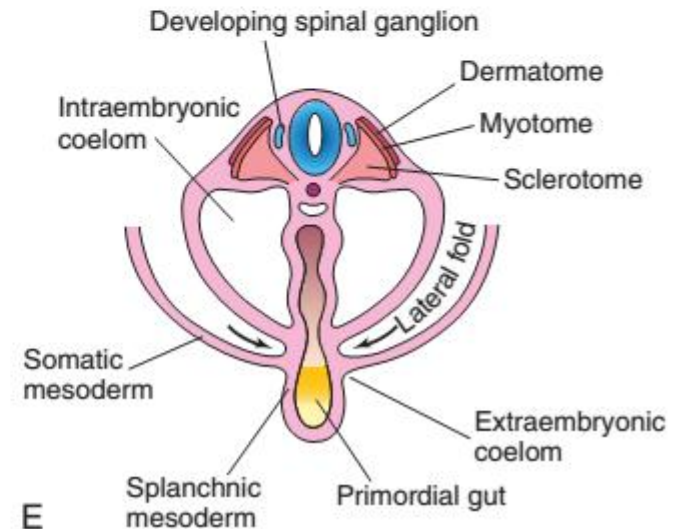
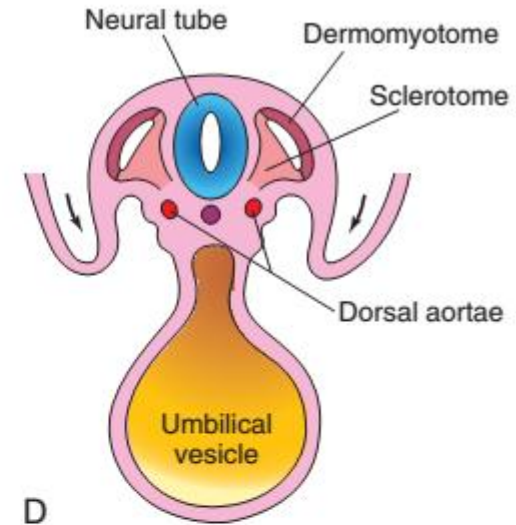
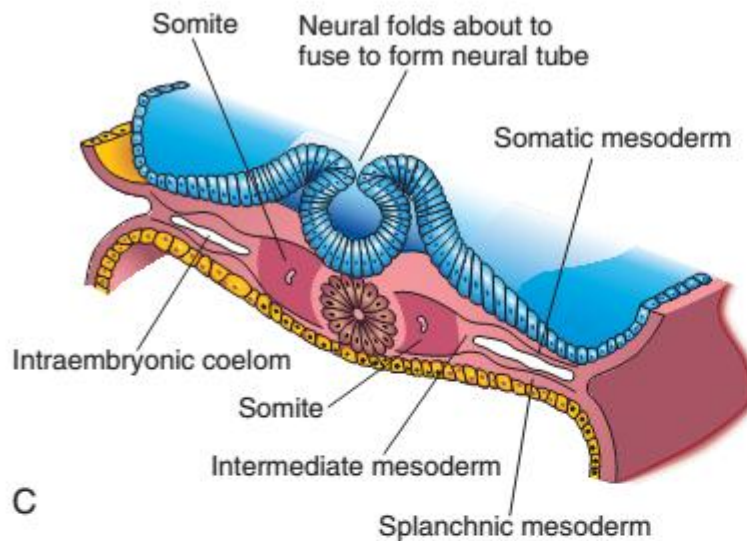
Somites

- **Paraxial mesoderm** differentiates and *begins to divide into cuboidal bodies* called **somites** by the end of 3rd week
- These blocks of mesoderm are **located on each side of developing neural tube**
- About 38 pairs of somites form during the somite period of human development (20-30 days)

Somites

- Each **somite differentiates** into **two parts**:
 - The **ventromedial part** is **sclerotome**
 - Its cells form the **vertebrae and ribs**
 - The **dorsolateral part** is the **dermomyotome**
 - Cells from **myotome** form myoblasts
 - Cells from **dermatome** form the dermis (fibroblasts)

Development of Skeletal system



Development of Bone and Cartilage

- At the end of the fourth week, the **sclerotome** cells form **mesenchyme**
- Bones first appear as condensations of mesenchymal cell
- **Two kinds of osteogenesis**
 - **Intramembranous bone formation**
 - **Most flat bones** develop
 - **Endochondral bone formation**
 - Mesenchymal models of **most limb bones** are transformed into cartilage bone models

Histogenesis of Cartilage

- Cartilage development begins from mesenchyme during the fifth week.
- In areas where cartilage is to develop, the mesenchyme condenses to form **chondrification centers**
- The mesenchymal cells differentiate first into prechondrocytes and then into **chondroblasts**
 - **chondroblasts** secrete collagenous fibrils and ground substance

Histogenesis of Cartilage

- **Three types of cartilage** are distinguished according to the type of matrix that is formed
 - **Hyaline cartilage**, the most widely distributed type (e.g., in joints)
 - **Fibrocartilage** (e.g., intervertebral discs)
 - **Elastic cartilage** (e.g., auricles of the external ears)

Histogenesis of Bone

- Bone primarily develops from two types of connective tissue
 - Mesenchyme
 - Cartilage
- bone consists of cells and an organic **bone matrix**
- **Two kinds of ossification**
 - **Intramembranous ossification**
 - **Endochondral ossification**

Intramembranous Ossification

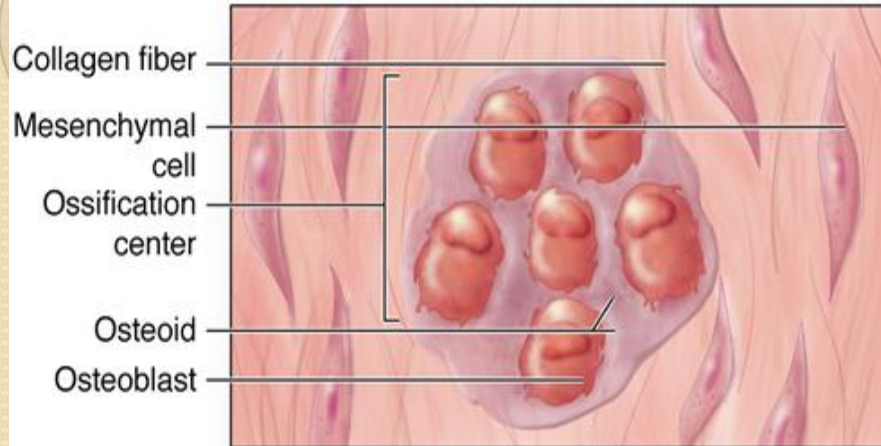
- occurs in mesenchyme that has formed a membranous sheath
- mesenchyme condenses and becomes highly vascular
- precursor cells differentiate into **osteoblasts**
- **Osteoblasts** deposit unmineralized matrix—**osteoid**
- Calcium phosphate is then deposited in the osteoid tissue

Intramembranous Ossification

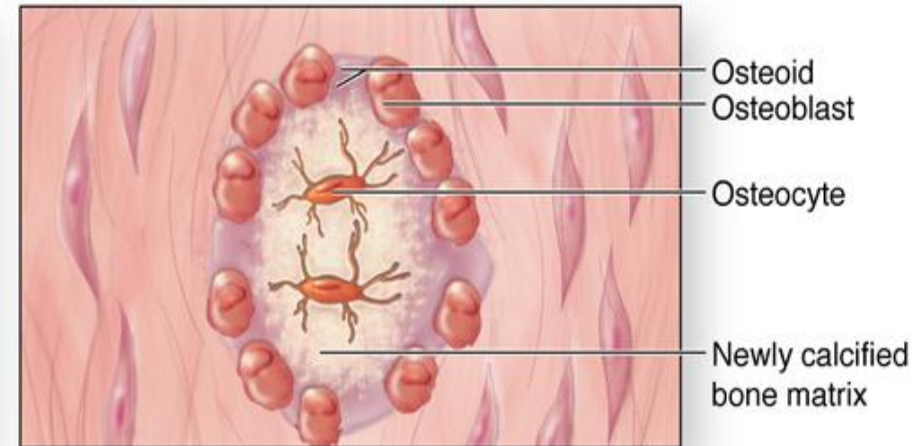


Intramembranous Ossification

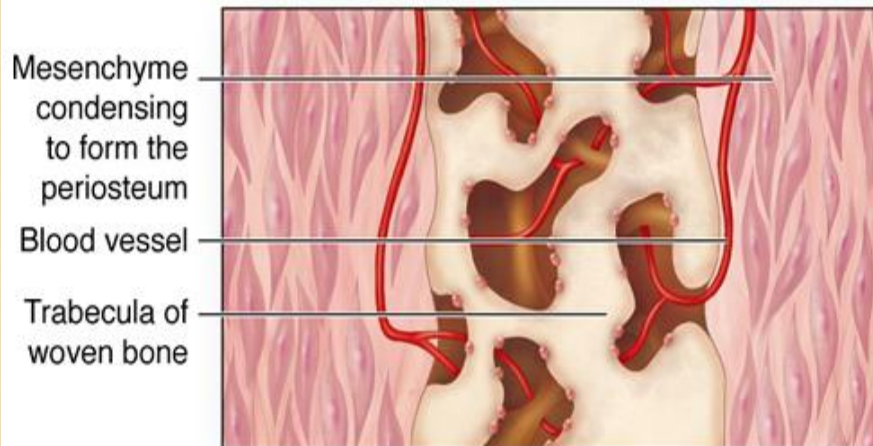
- a** Ossification centers form within thickened regions of mesenchyme.



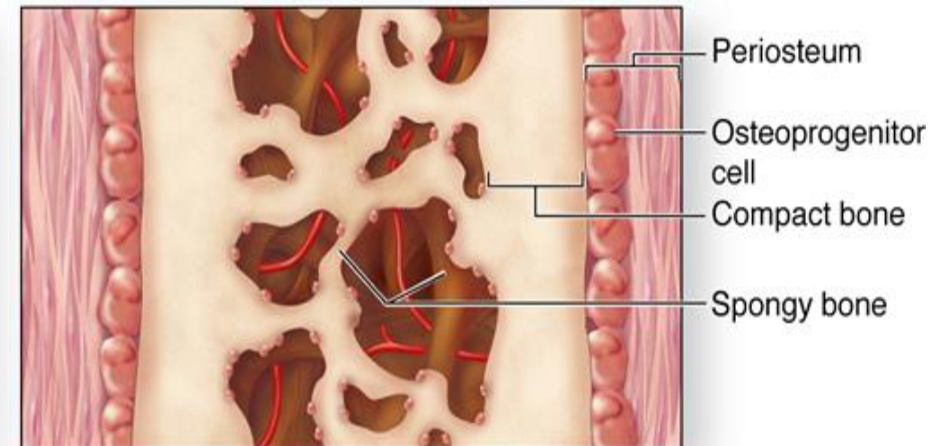
- b** Osteoid undergoes calcification.



- c** Woven bone and surrounding periosteum form.



- d** Lamellar bone replaces woven bone, as compact and spongy bone form.



Intramembranous Ossification

- Concentric lamellae develop around blood vessels, forming **osteons**.
- Some osteoblasts remain at the periphery of the bone forming plates of compact bone on the surfaces.
- Between the surface plates, the intervening bone remains spongy.
- This spongy environment is somewhat accentuated by the action of **osteoclasts**, which reabsorb bone.
- In between the spongy bone, the mesenchyme differentiates into **bone marrow**.

Endochondral Ossification

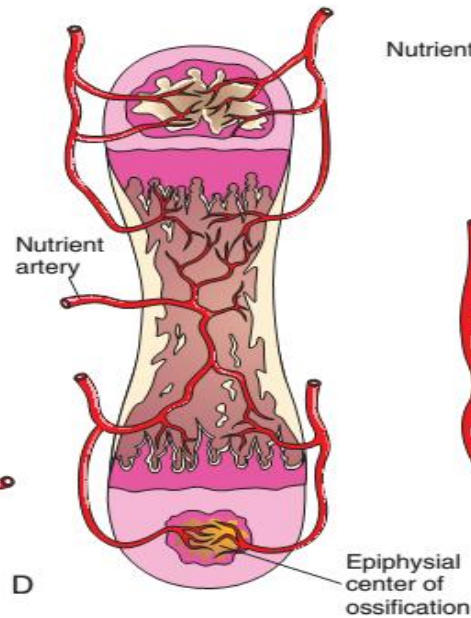
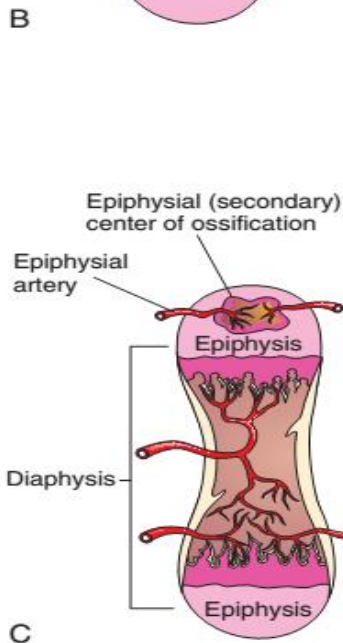
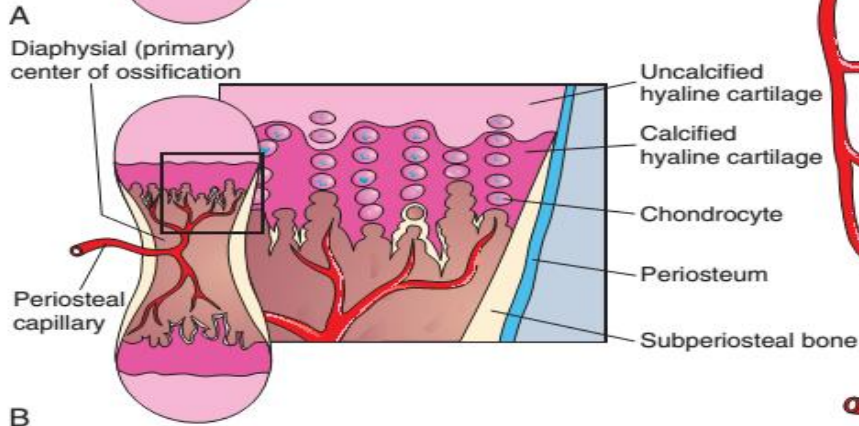
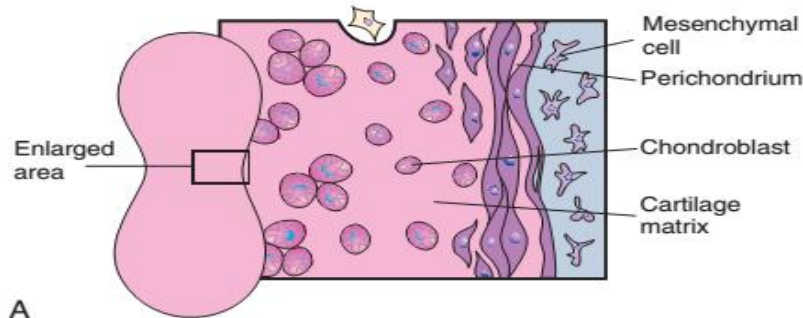
- Occurs in **preexisting cartilaginous models**
- **Steps:**
 - **primary center of ossification** appears in the diaphysis
 - At this center of ossification, chondrocytes increase in size
 - matrix becomes calcified, and the cells die.
 - Concurrently, a thin layer of bone is deposited under the perichondrium
 - perichondrium becomes the periosteum.

Endochondral Ossification

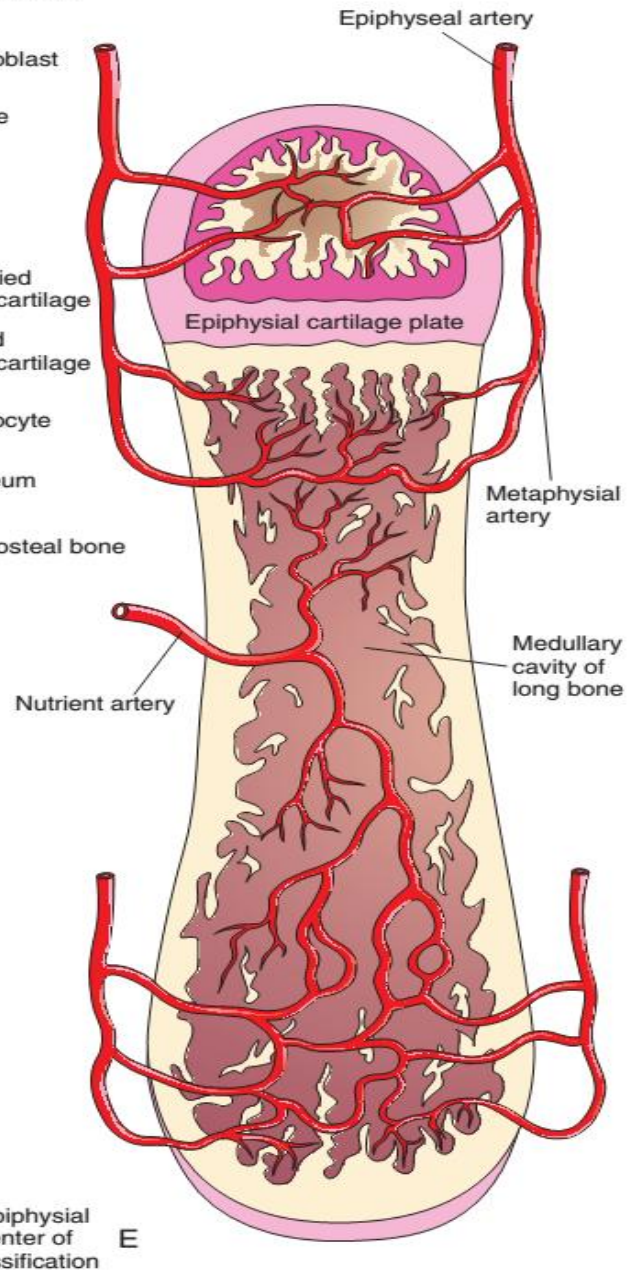
- Steps:
 - Invasion by vascular connective tissue
 - Some invading cells differentiate into hemopoietic cells
 - This process continues toward the epiphyses
 - bone remodeling by the action of osteoclasts and osteoblasts.

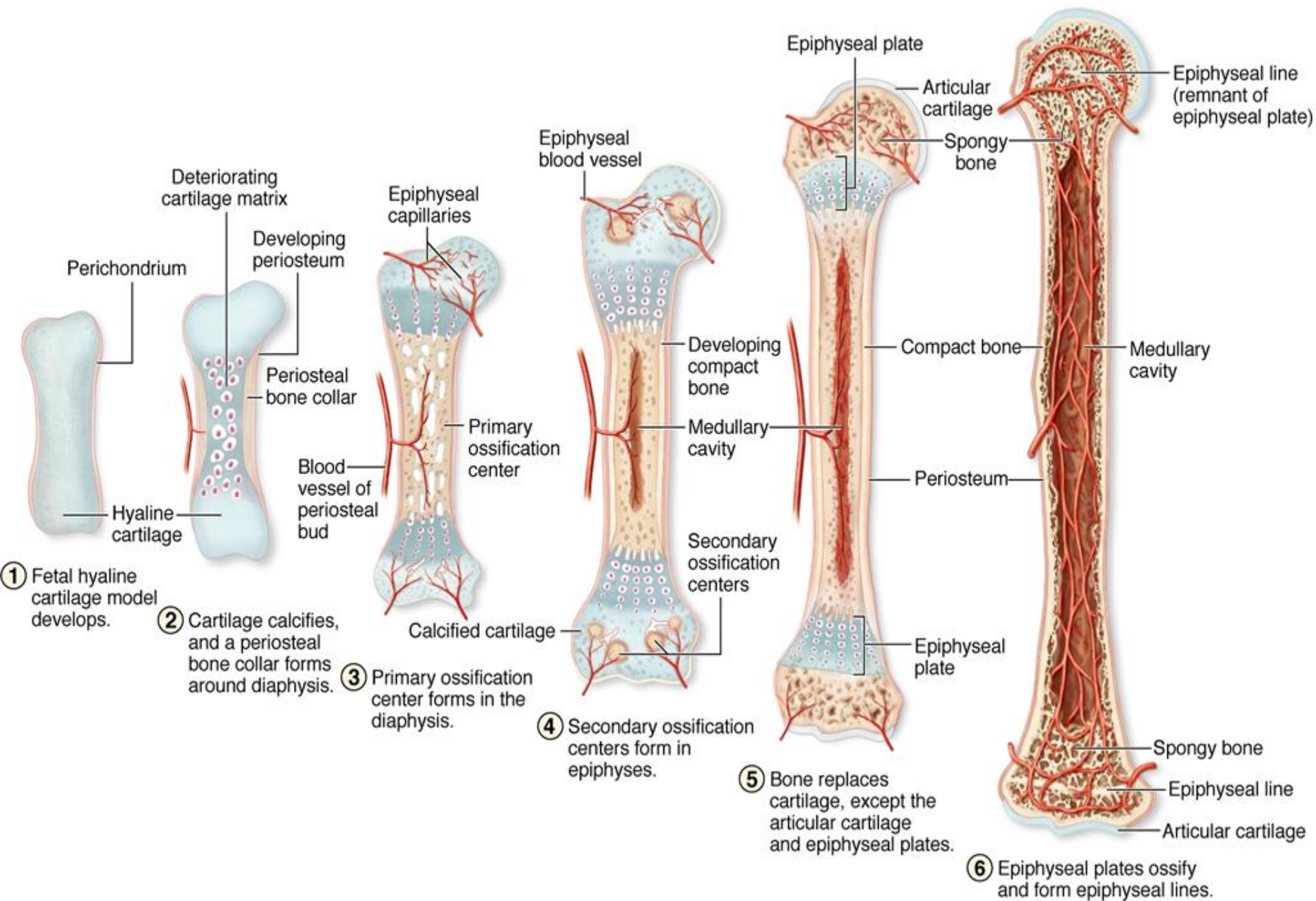
Cartilage Model of Bone

Cartilage
 Calcified cartilage
 Bone
 Arteries



Intracartilaginous Ossification





Endochondral Ossification

- Lengthening of long bones
 - occurs at the diaphysial-epiphysial junction.
 - It depends on the epiphysial cartilage plates (growth plates)
 - Cartilage cells in this region proliferate by mitosis.
- Toward the diaphysis
 - cartilage cells hypertrophy
 - matrix becomes calcified.

Endochondral Ossification

- **Spicules**

- Isolated from each other by vascular invasion from medullary cavity.
- Bone is deposited within it by osteoblasts

- **Resorption**

- keeps the spongy bone masses relatively constant in length
- Enlarges the medullary cavity.

Endochondral Ossification

- **Ossification of limb bones**
 - begins at the end of the embryonic period
 - It is dependent on maternal supply of calcium and phosphorus.
- **At birth**
 - diaphyses are largely ossified
 - but most of the epiphyses are still cartilaginous.

Endochondral Ossification

- **Secondary ossification centers**
 - Appear in the epiphyses during the first few years after birth.
 - Epiphyseal cartilage cells hypertrophy, and there is invasion by vascular connective tissue.
- After secondary ossification the following structure remain cartilaginous
 - articular cartilage
 - epiphyseal cartilage plate
- Upon completion of growth
 - epiphyseal cartilage plate will be replaced by spongy bone
 - epiphyses and diaphysis unite

Endochondral Ossification

- Growth in the diameter of a bone results from
 - deposition of bone at the periosteum
 - resorption on the internal medullary surface.
- Rate of deposition and resorption is balanced to regulate:
 - thickness of the compact bone
 - size of the medullary cavity

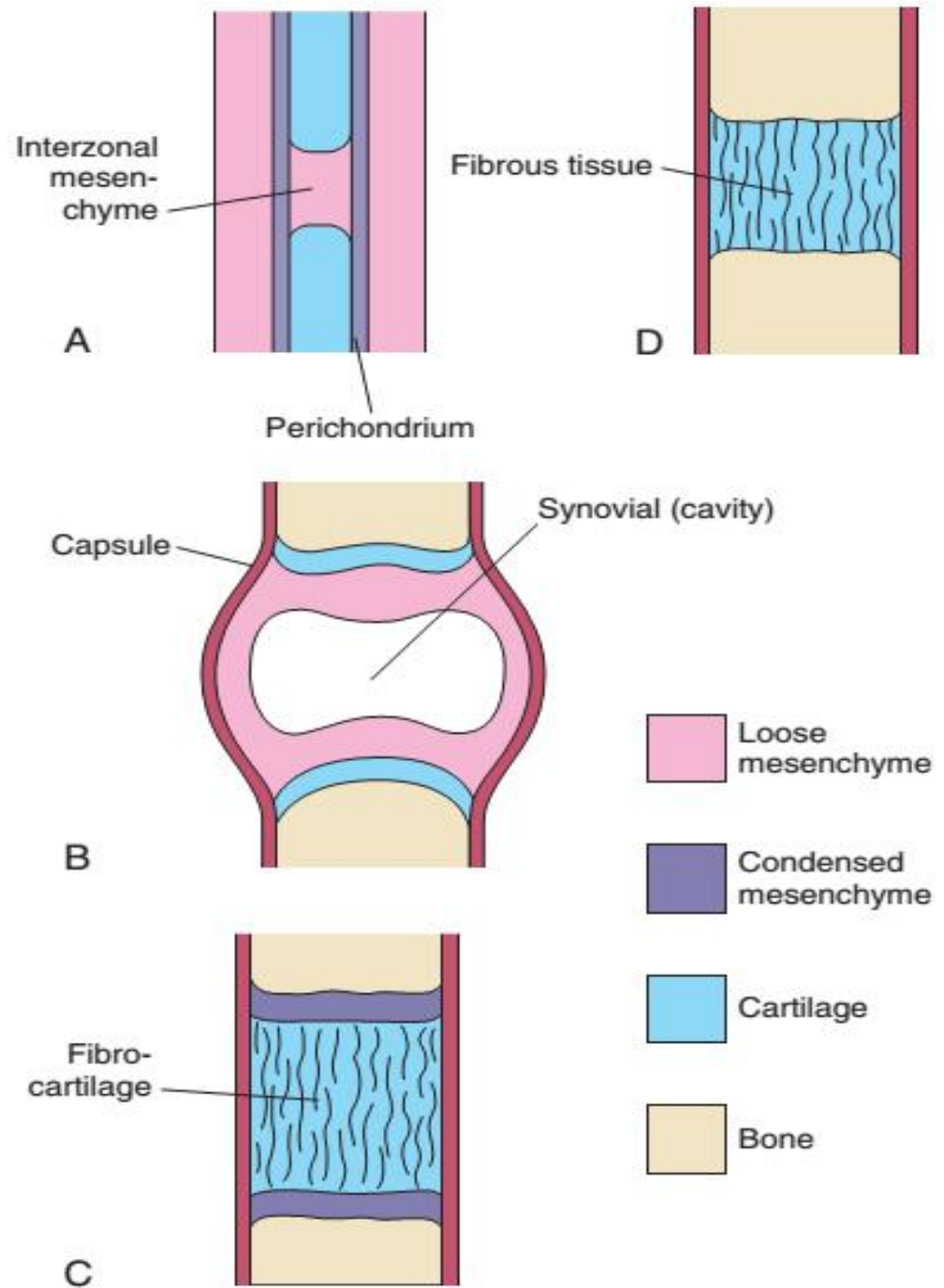
Endochondral Ossification

- **Development of irregular bones**
 - Is similar to that of the epiphyses of long bones.
 - Ossification begins centrally and spreads in all directions.

Development of Joints

- begin to develop with the appearance of **condensed mesenchyme** in the joint interzone during the sixth week
- resemble adult joints by the end of the eighth week
- Joints are classified as
 - **Fibrous joints**
 - **Cartilaginous joints**
 - **Synovial joints**

A-Development of joints during the sixth and seventh weeks
B-synovial joint
C-cartilaginous joint
D-fibrous joint



Development of Joints

- **Fibrous Joints**

- **interzonal mesenchyme** differentiates into dense fibrous tissue
- E.g. sutures

- **Cartilaginous Joints**

- interzonal mesenchyme differentiates into
 - **hyaline cartilage** (e.g., costochondral joints)
 - **fibrocartilage** (e.g., pubic symphysis)

Synovial Joints

- interzonal mesenchyme differentiates as follows:
 - Peripherally, it forms the **joint capsule and other ligaments**.
 - Centrally it disappears and the resulting space becomes the **joint cavity** or **synovial cavity**.
 - Where it lines the joint capsule and articular surfaces, it forms the **synovial membrane**, which **secretes the synovial fluid**.



Development of Axial Skeleton

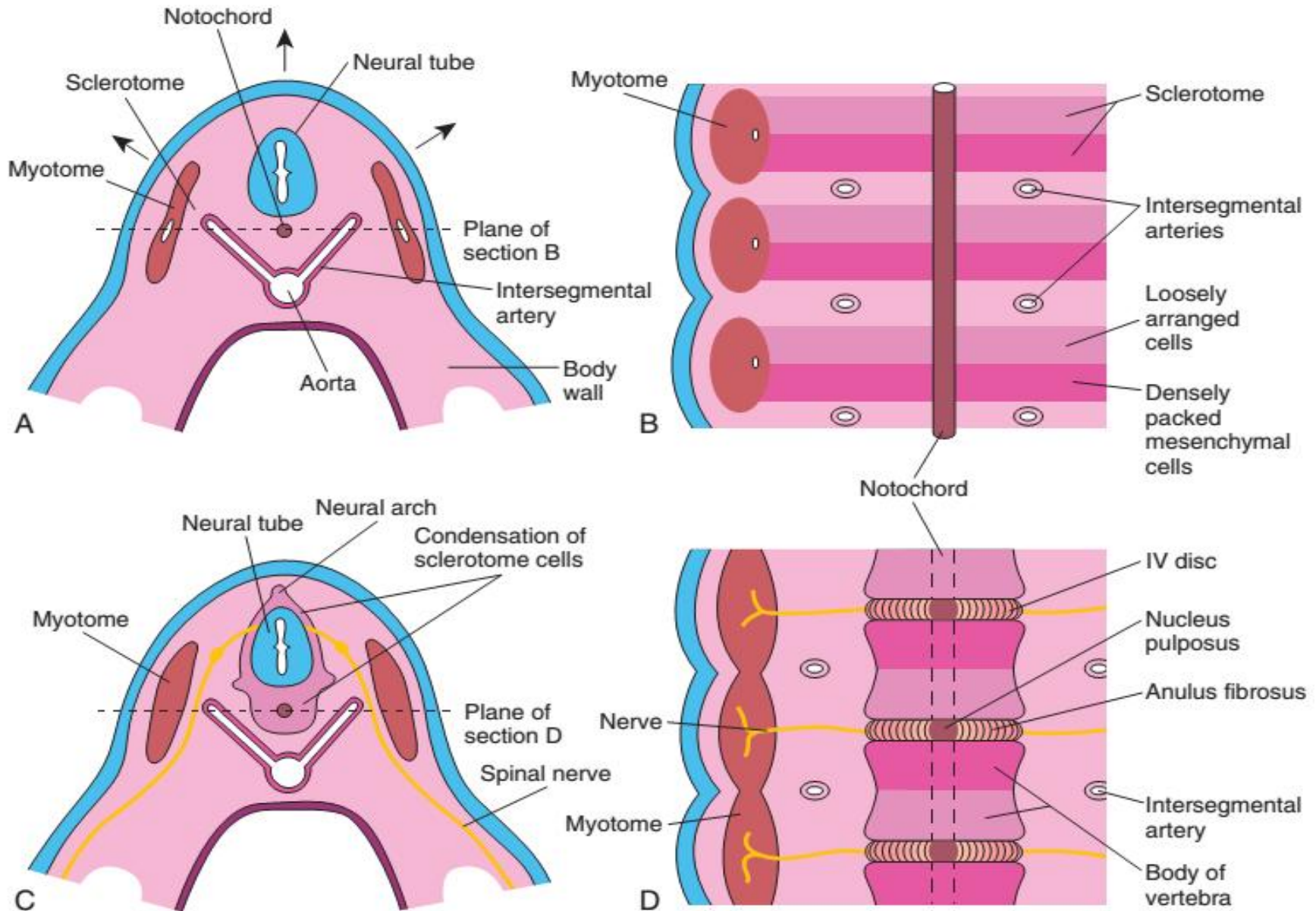
Development of Vertebral Column

- Vertebrae form from the **sclerotome** portions of the somites
- A typical vertebra consists of
 - **Vertebral arch and foramen**
 - **Body**
 - **Transverse processes**
 - **Spinous process**
- During the fourth week, cells in the **sclerotomes** migrate to and surround the **neural tube (Spinal cord)**

Development of Vertebral Column

- Each sclerotome consists of
 - *loosely arranged* cells **cranially**
 - *densely packed* cells **caudally**
- The **caudal half** of a sclerotome **fuses** with the **cranial half of the sclerotome** below to form the **centrum** or **vertebral body**
- The mesenchymal cells that surround the neural tube form the **neural arch or vertebral arch**
- The mesenchymal cells in the body wall form the **costal processes**, which form the ribs in the thoracic region
- The vertebral arches of each vertebra join dorsally to form the **spinous processes**

Development of Vertebral Column



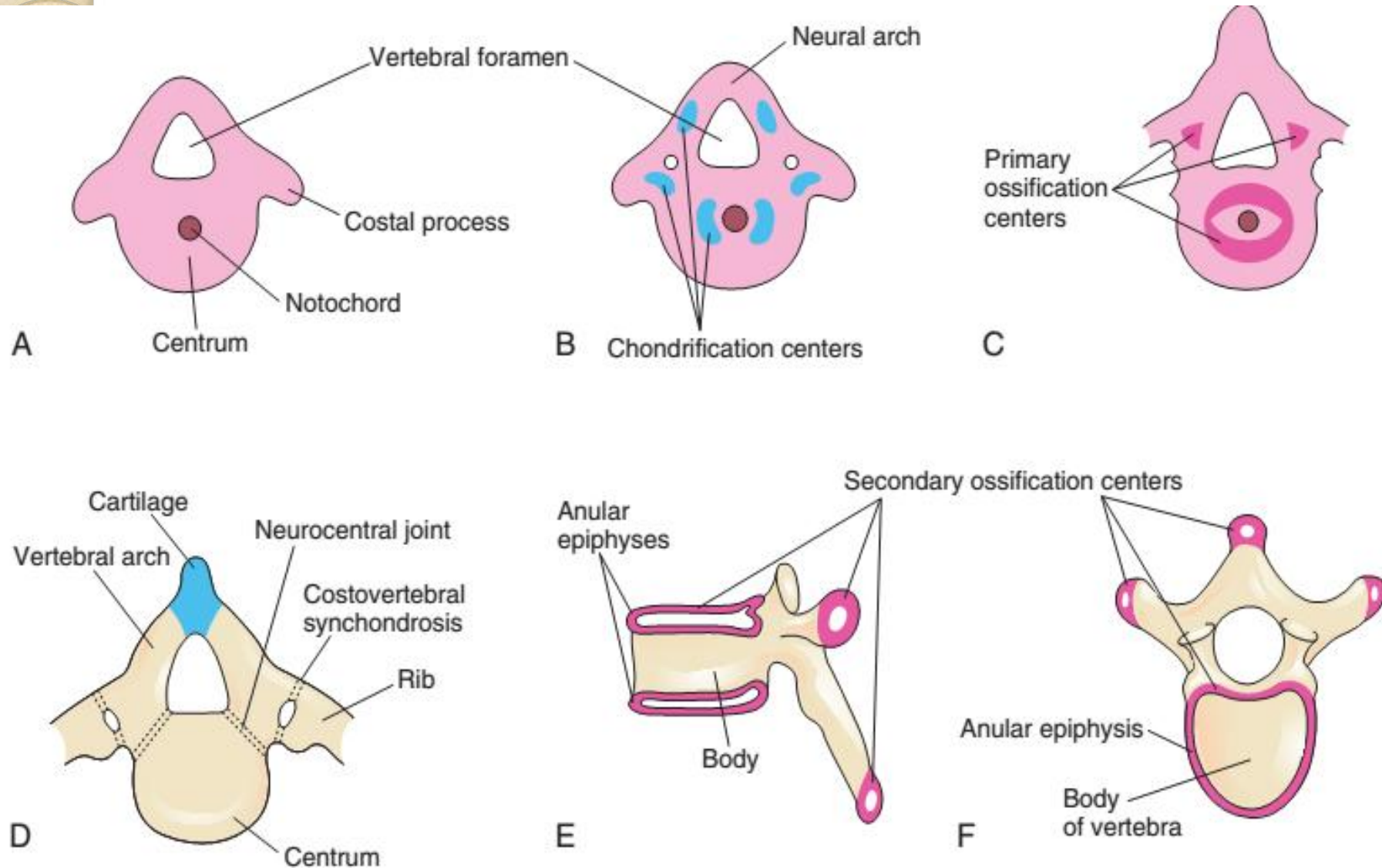
Development of Vertebral Column

- The **intervertebral disc** has
 - **Annulus fibrosus** – outer circularly arranged collagenous fibers
 - *develops from the densely packed lower portion of the sclerotome*
 - **Nucleus pulposus** – central gelatinous core
 - *derived from the expansion of notochord between the vertebrae*
- The **rest of the notochord** at the level of the vertebral bodies **soon disappears**

Cartilaginous Stage of Vertebral Development

- During the sixth week, **chondrification centers** appear in each mesenchymal vertebra
- At the end of the embryonic period, the two centers in each centrum **fuse** to **form a cartilaginous centrum**
- Concomitantly, the centers in the neural arches fuse with each other and the centrum.
- The **spinous** and **transverse processes** develop from extensions of chondrification centers in the **vertebral arch**.

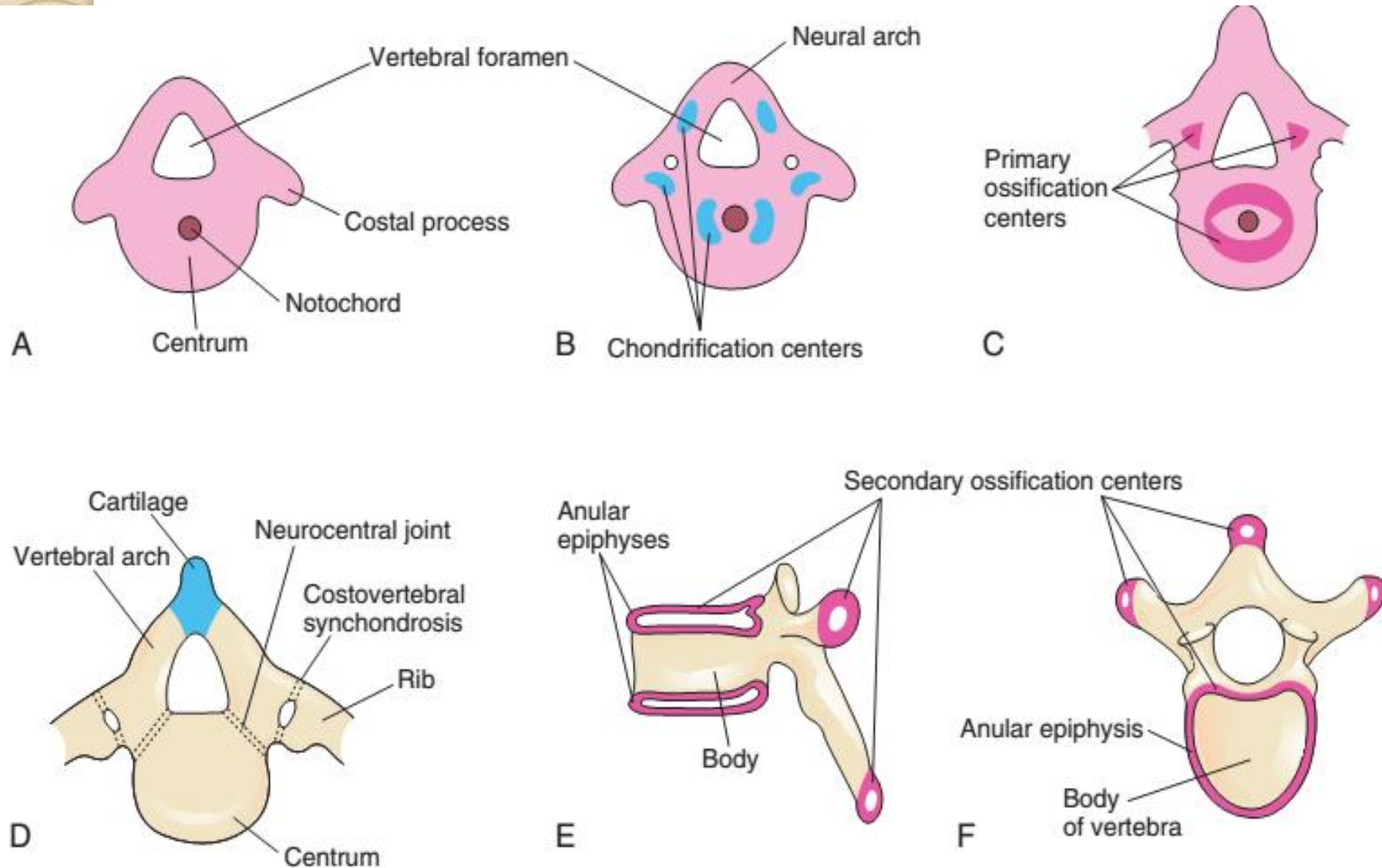
Stages of vertebral development



Bony Stage of Vertebral Development

- **Ossification** of typical vertebrae *begins during the embryonic period*
- It usually **ends by the twenty-fifth year**
- There are **two primary ossification centers**
 - **ventral** and **dorsal** for the centrum
- These primary ossification centers soon **fuse to form one center**

Stages of vertebral development



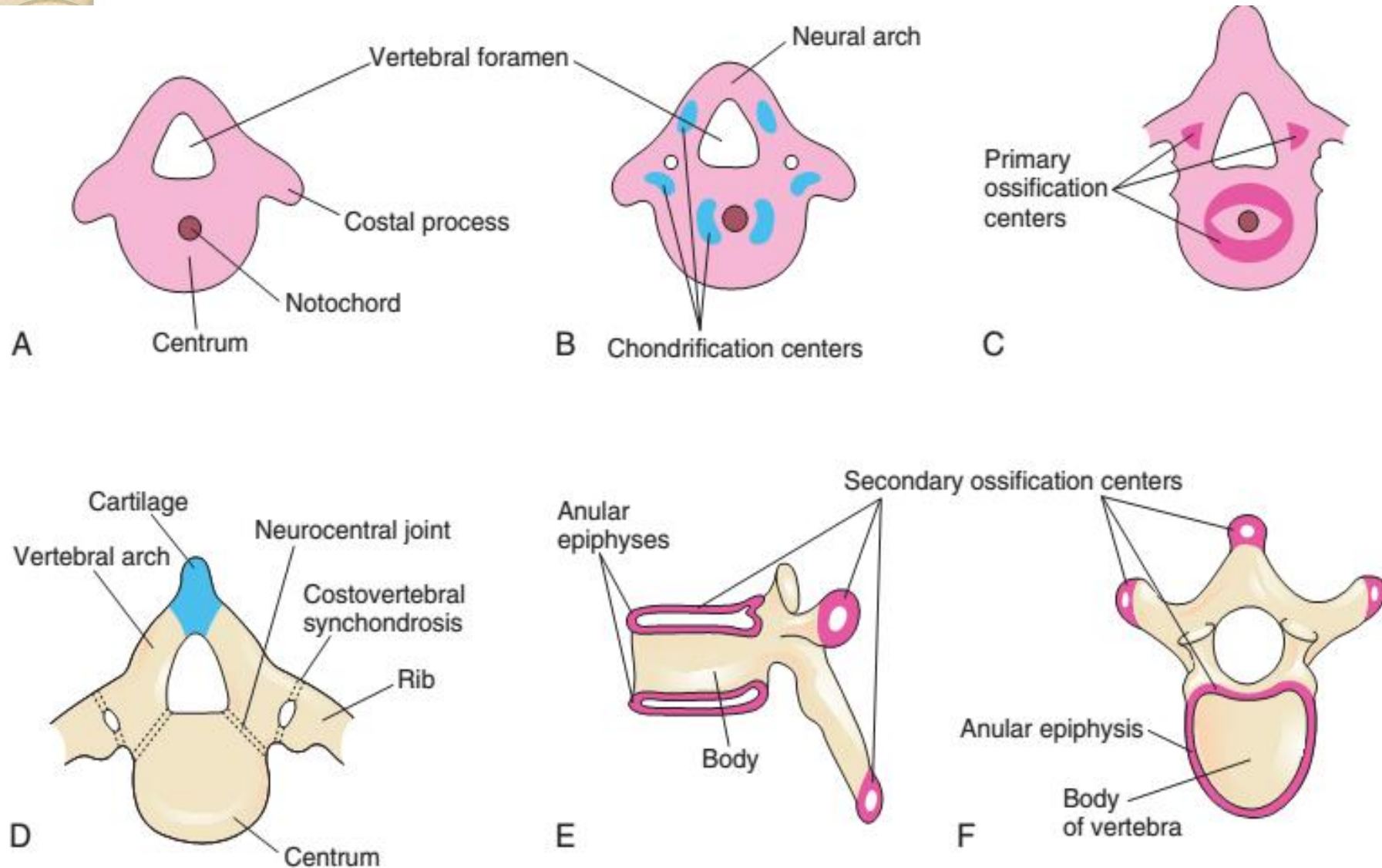
Bony Stage of Vertebral Development

- **Three primary** centers are present by the end of the embryonic period:
 - **One in the centrum**
 - **One in each half of the vertebral arch**
- *Ossification becomes evident in the vertebral arches during the **eighth week***

Bony Stage of Vertebral Development

- At birth each vertebra consists of three bony parts connected by cartilage
- The bony halves of the vertebral arch usually fuse during the first 3 to 5 years
- **The arches first unite in the lumbar region**
- **This union progresses cranially**
- The vertebral arch articulates with the centrum at cartilaginous neurocentral joints

Stages of vertebral development



Bony Stage of Vertebral Development

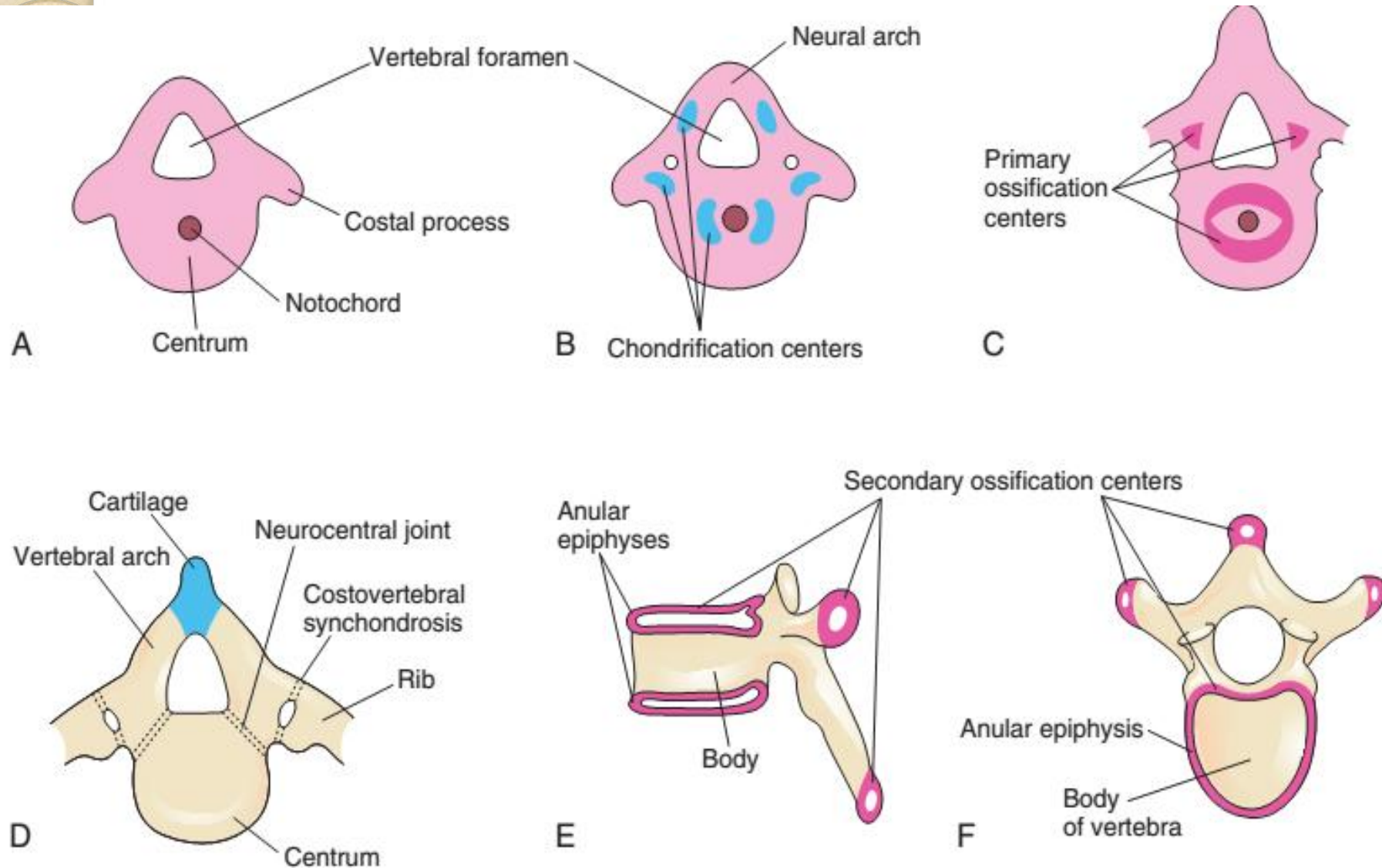
- These articulations permit the vertebral arches to grow as the spinal cord enlarges
- These joints disappear when the vertebral arch fuses with the centrum during the third to sixth years
- The vertebral body is a composite of the anular epiphyses and the mass of bone between them

Bony Stage of Vertebral Development

Five secondary ossification centers appear in the vertebrae after puberty:

- One for the tip of the spinous process
- One for the tip of each transverse process
- Two anular epiphysis, one on the superior and one on the inferior rim of the vertebral body

Stages of vertebral development



Bony Stage of Vertebral Development

- The vertebral body includes the centrum, parts of the vertebral arch, and the facets for the heads of the ribs
- All secondary centers unite with the rest of the vertebra around 25 years of age
- Exceptions to the typical ossification of vertebrae occur in the atlas, axis, C7, lumbar vertebrae, sacrum and coccyx

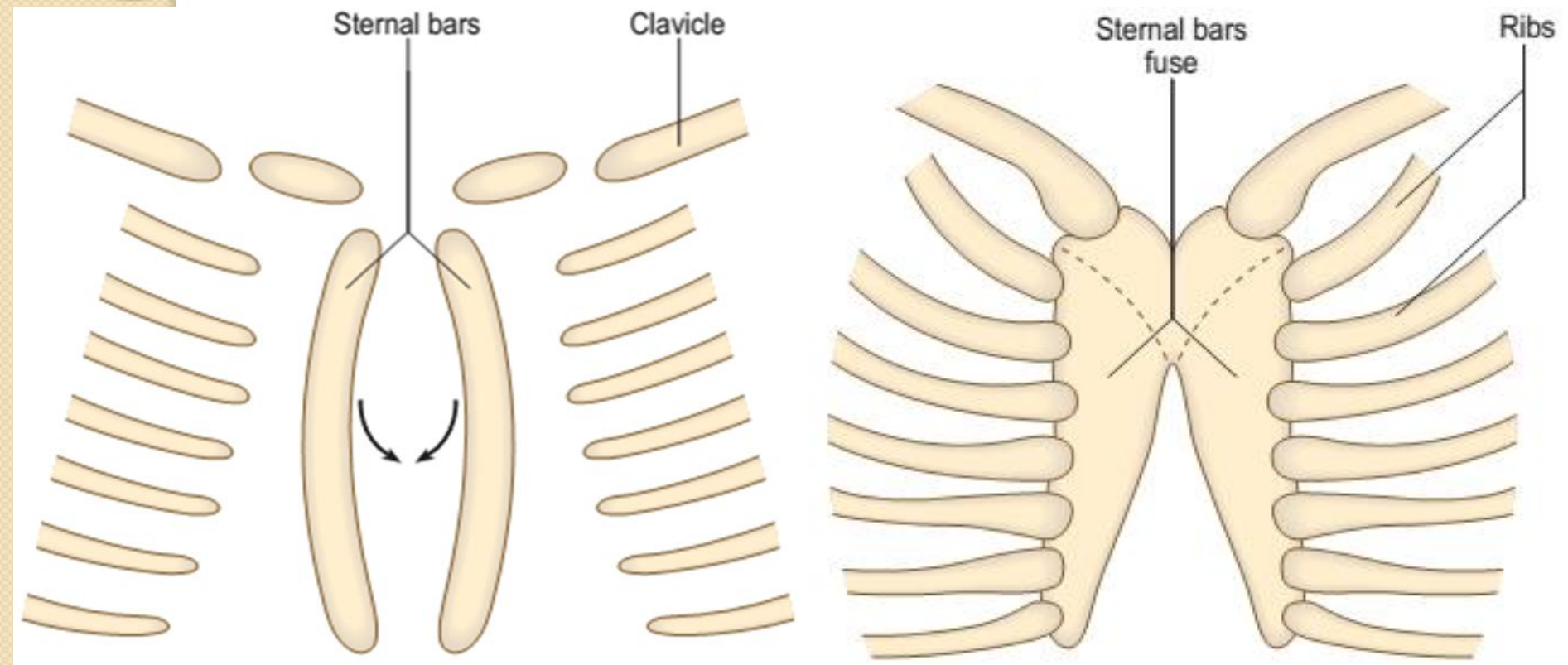
Development of Ribs

- develop from the mesenchymal **costal processes** of the thoracic vertebrae
- These become cartilaginous, grow laterally toward the sternum, and become ossified
- They become cartilaginous during the embryonic period and ossify during the fetal period.

Development of Sternum

- develops from a pair of cartilaginous bars (**sternal bars**) that form in the ventral body wall.
- These bars fuse in the midline in a craniocaudal sequence to form the manubrium, the body of the sternum and xiphoid process

Development of Ribs and Sternum



Development of Skull

- The **Skull** develops from mesenchyme around the developing brain
- The **Skull** consists of:
 - **Neurocranium**, a *protective case for the brain*
 - **Viscerocranium**, the *skeleton of the face*

Neurocranium

- The neurocranium has two portions:
 - **membranous part**, consisting of **flat bones**, which surround the brain as a vault
 - **cartilaginous part**, or **chondrocranium**, which forms bones of the **base of the skull**

Cartilaginous Neurocranium

- Initially the cartilaginous neurocranium or chondrocranium consists of the cartilaginous base of the developing cranium
- It forms by the fusion of several cartilages
- Later, endochondral ossification of the chondrocranium forms the bones in the base of the cranium

Cartilaginous Neurocranium

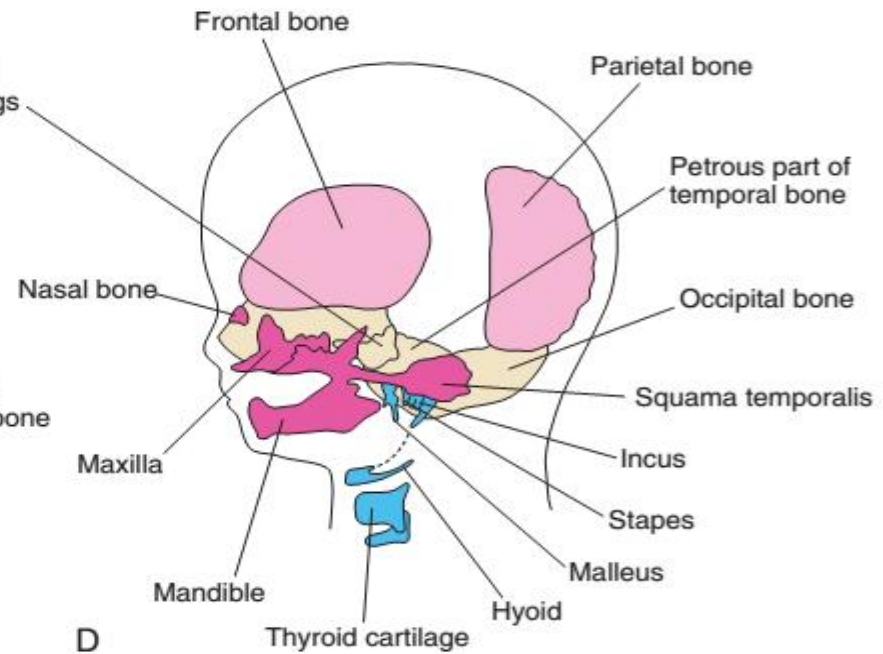
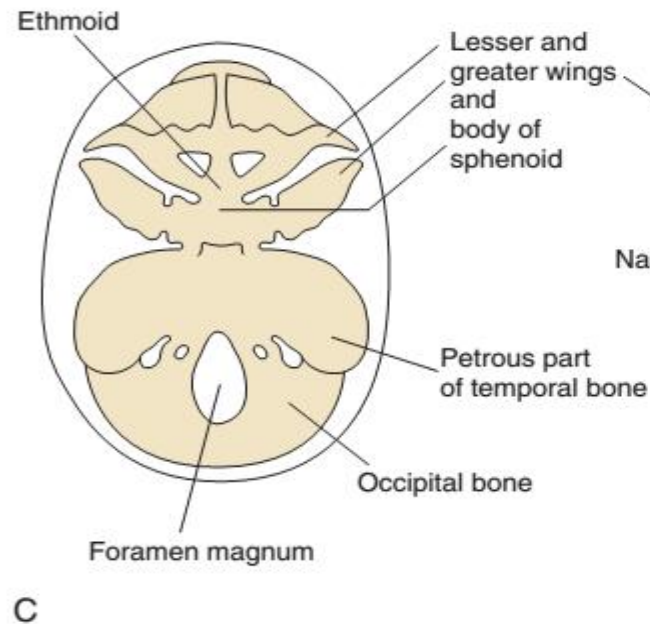
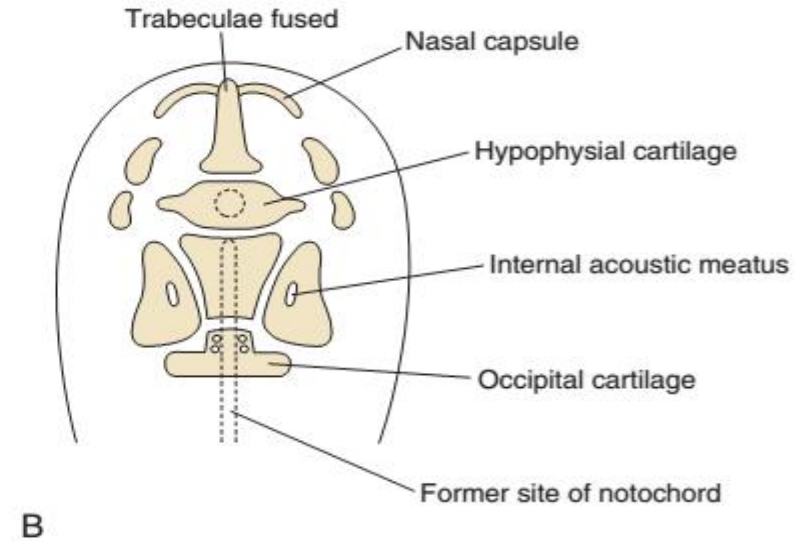
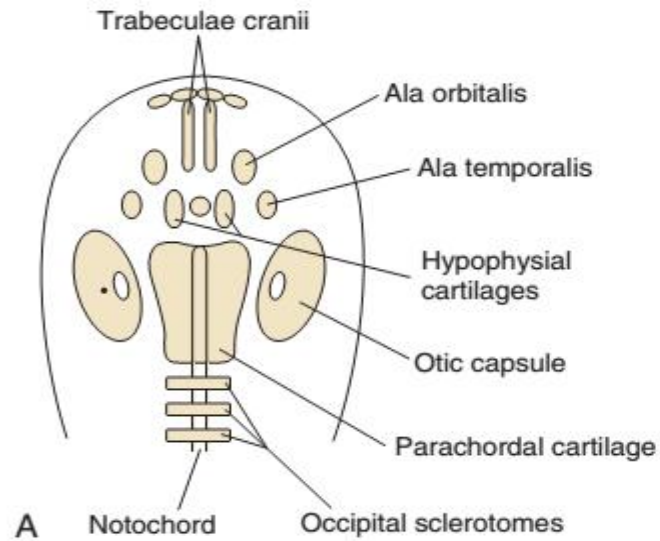
- The ossification pattern of these bones beginning with occipital bone, body of sphenoid, and ethmoid bone
- The parachordal cartilage or basal plate forms around the cranial end of the notochord
- It fuses with the cartilages derived from the sclerotome regions of the occipital somites

Cartilaginous neurocranium

Membranous neurocranium

Cartilaginous viscerocranium

Membranous viscerocranium



Cartilaginous Neurocranium

- This cartilaginous mass contributes to the base of the occipital bone
- Later extensions grow around the cranial end of the spinal cord
- These extensions form the boundaries of the foramen magnum

Cartilaginous Neurocranium

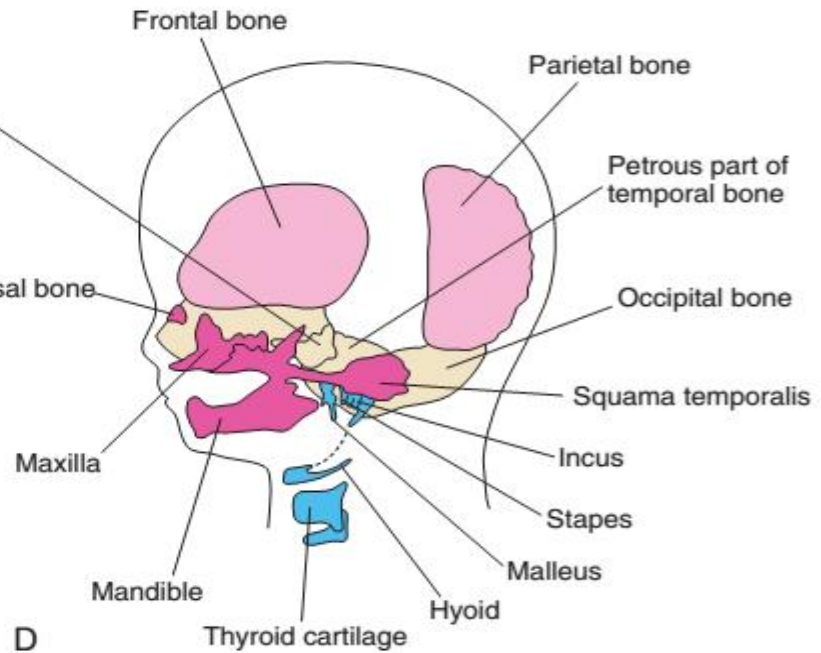
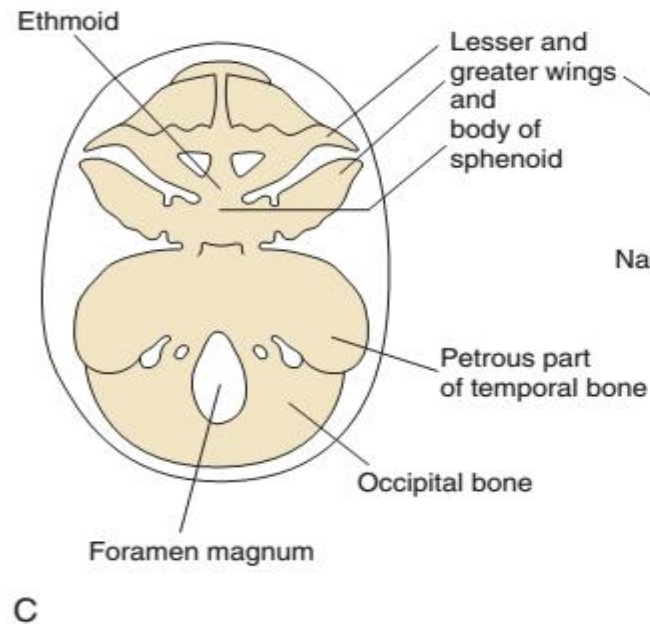
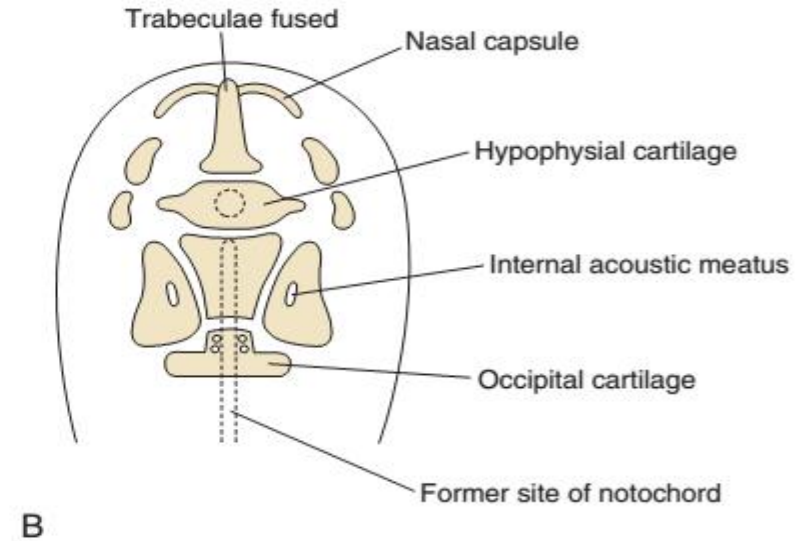
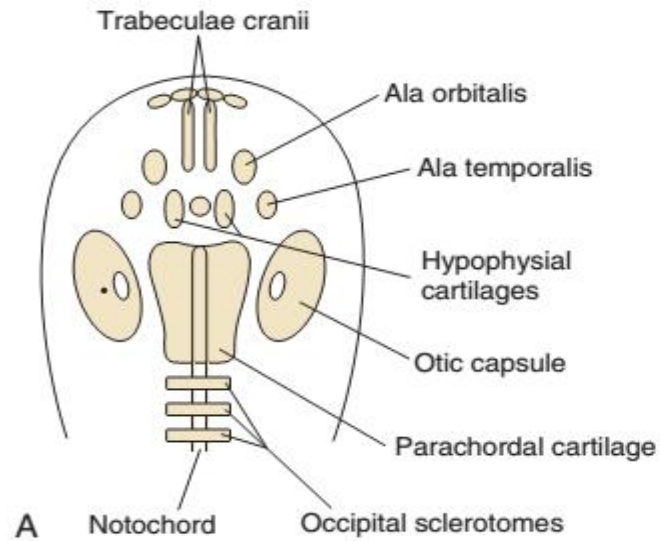
- Hypophysial cartilage forms around the developing pituitary gland
- It fused to form the body of the sphenoid bone
- The trabeculae cranii fuse to form the body of the ethmoid bone
- The ala orbitalis forms the lesser wing of the sphenoid bone

Cartilaginous neurocranium

Membranous neurocranium

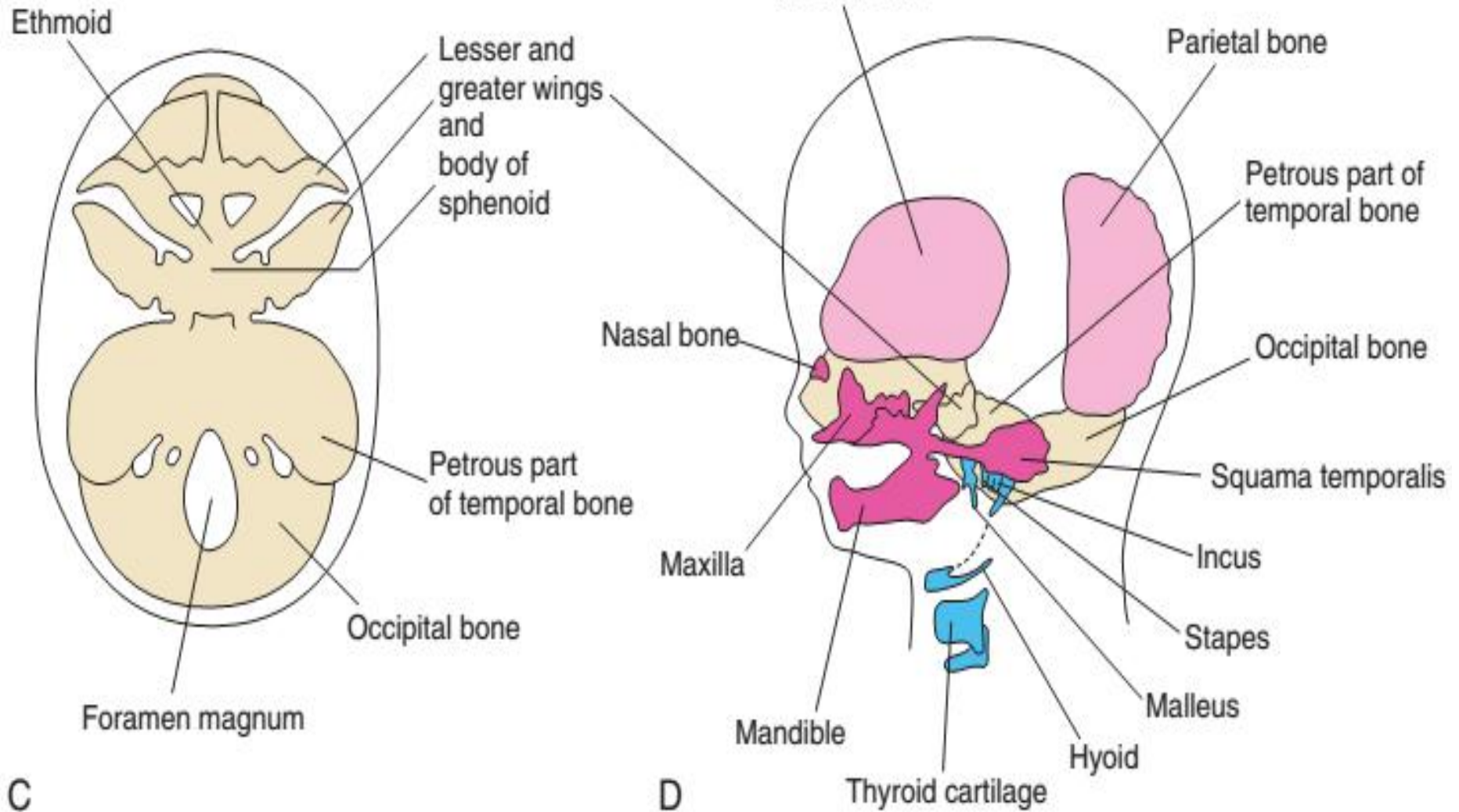
Cartilaginous viscerocranium

Membranous viscerocranium



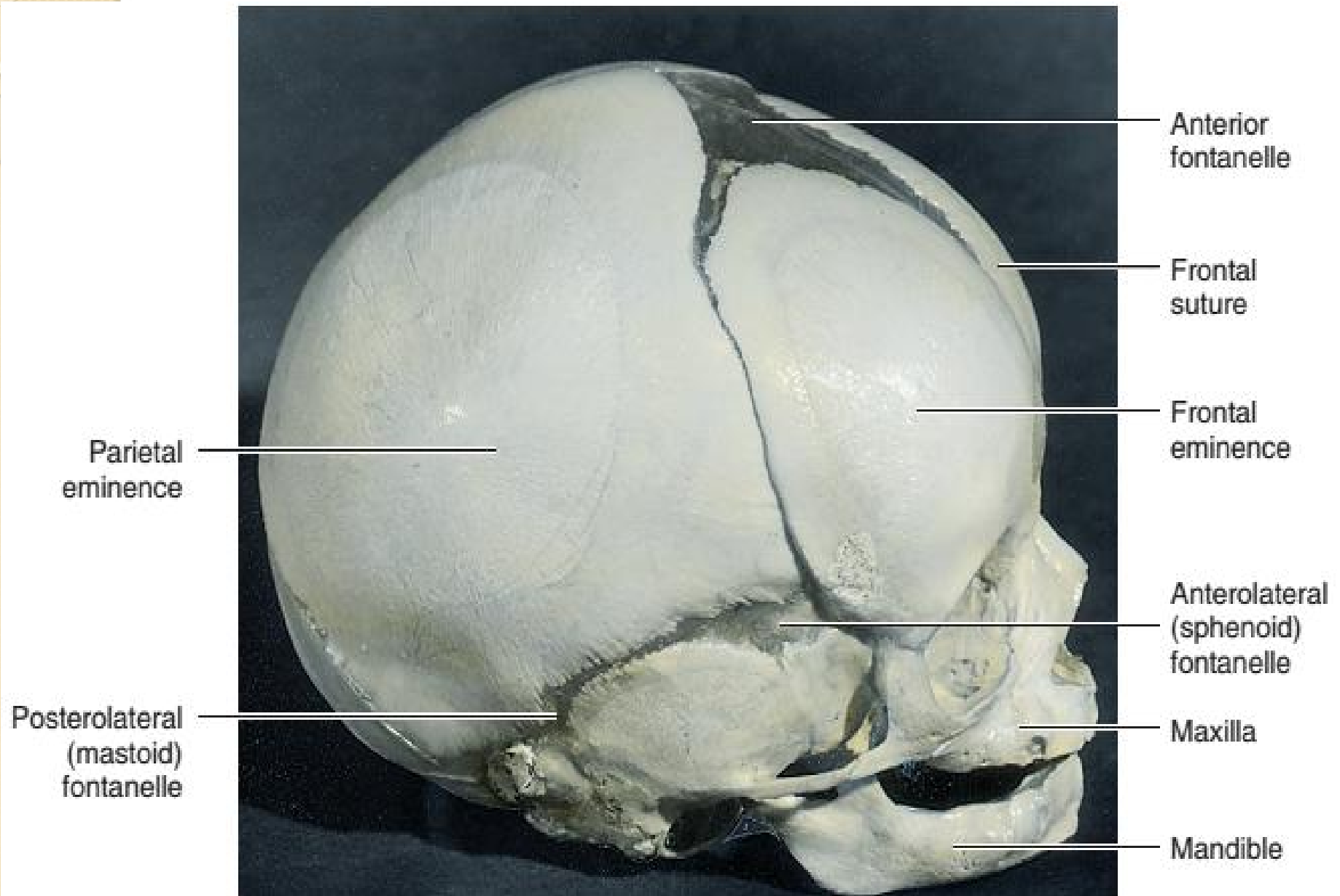
Cartilaginous Neurocranium

- Otic capsules develop around the otic vesicles, the primordia of the internal ears
- They form the petrous and mastoid parts of the temporal bone
- Nasal capsules develop around the nasal sacs
- They contribute to the formation of the ethmoid bone



Membranous Neurocranium

- Intramembranous ossification occurs in the mesenchyme at the sides and top of the brain forming calvaria (cranial vault)
- During fetal life the flat bones of the calvaria are separated by **dense connective tissue membranes**, that form the **sutures**
- Six large fibrous areas **fontanelles** are present where several sutures meet



Membranous Neurocranium

- The softness of bones and their loose connections at the sutures enable the calvaria to change shape during birth
- During molding of the fetal cranium, the frontal bones become flat
- The occipital bone is drawn out
- Parietal bone overrides the other one
- Shape of the calvaria returns to normal in few days after birth

Sagittal
suture

Lambdoid
suture

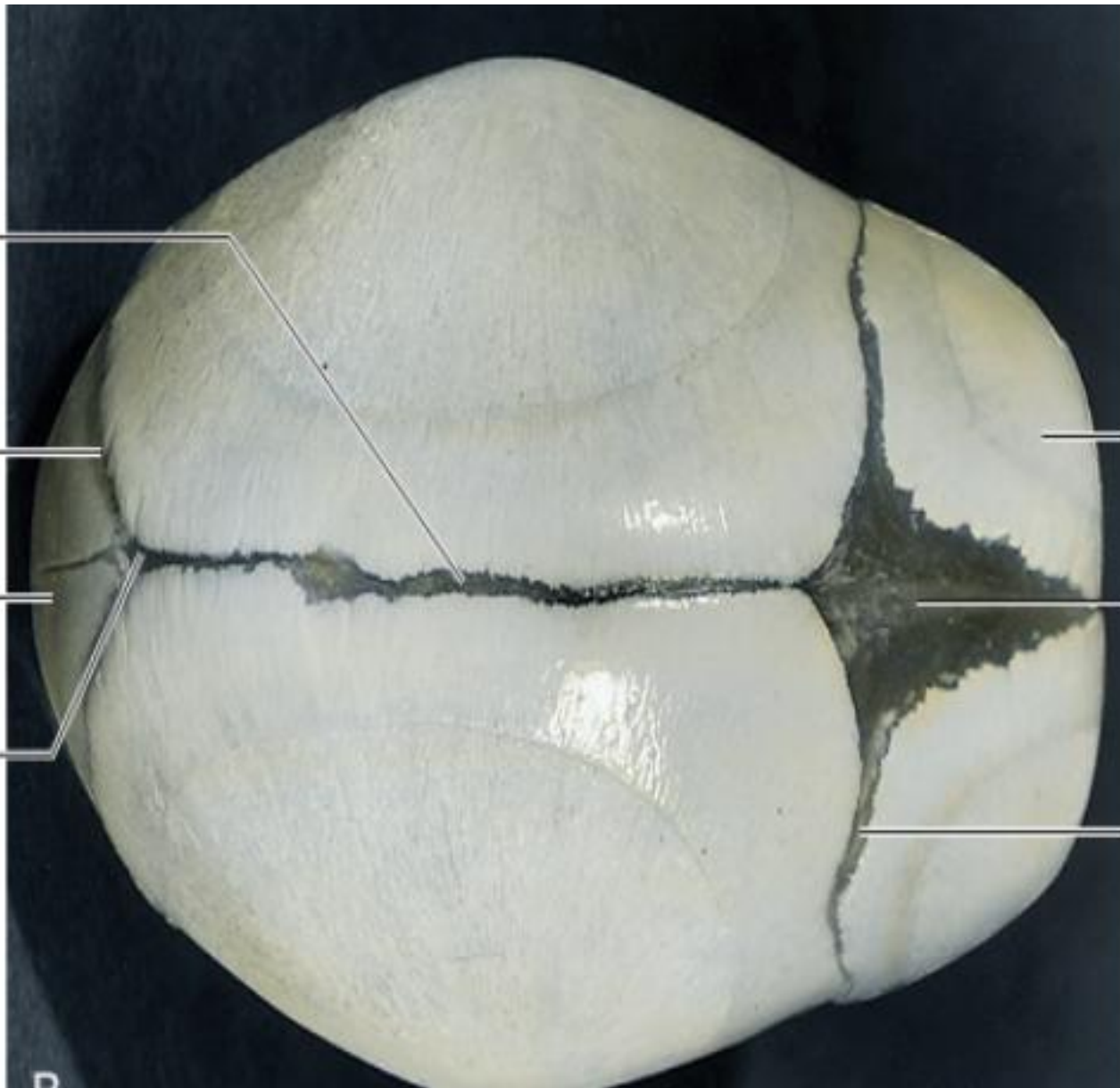
Occipital
bone

Posterior
fontanelle

Frontal
bone

Anterior
fontanelle

Coronal
suture



Cartilaginous Viscerocranium

- These parts of the fetal cranium are derived from the cartilaginous skeleton of the first two pairs of pharyngeal arches
 - **1st arch**: malleus and incus
 - **2nd arch**: stapes, styloid process, lesser cornu and body of hyoid bone
 - **3rd arch**: greater horn and lower part of hyoid bone
 - **4th to 6th arches**: laryngeal cartilages

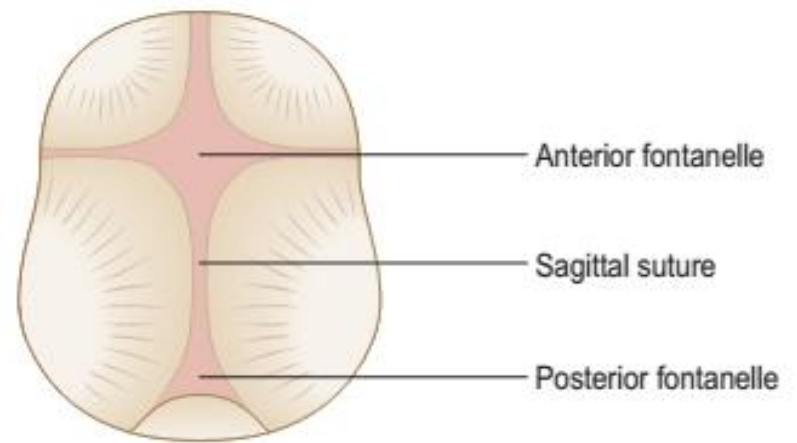
Membranous Viscerocranium

- Intramembranous ossification occurs in the **maxillary prominence** of the first pharyngeal arch
 - Subsequently forms the **squamous temporal, maxillary, and zygomatic bones**
 - The squamous temporal bones become part of the neurocranium
- **Mandibular prominence** undergoes **intramembranous ossification** to form **mandible**

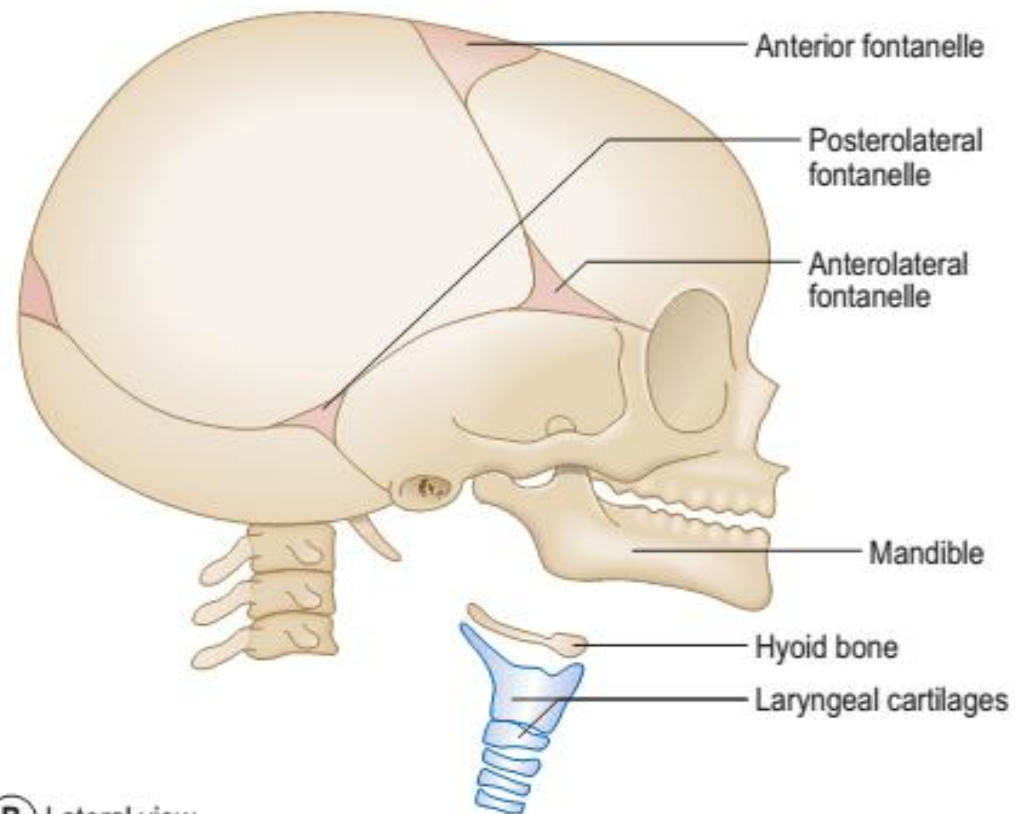
Newborn Cranium

- Newborn's cranium is round and thin
- It is large in proportion to the rest of the skeleton
- Face is relatively small compared with the calvaria
- The small facial region of cranium results from:
 - Small size of the jaw
 - Absence of paranasal air sinuses
 - Underdeveloped facial bones at birth

Fetal Skull



A Superior view



B Lateral view

Postnatal Growth of Cranium

- The fibrous sutures of the newborn's calvaria permit the brain to enlarge during infancy and childhood
- The increase in the size of the calvaria is greatest during the first 2 years
- This is the period of rapid postnatal growth of the brain
- Calvaria normally increases in capacity until about 16 years of age

Postnatal Growth of Cranium

- There is a rapid growth of the face and jaws coinciding with eruption of teeth
- These facial changes are more marked after the secondary teeth erupt
- Enlargement of frontal and facial regions also increase with increase in size of paranasal sinuses
- Most paranasal sinuses are rudimentary or absent at birth
- Growth of these sinuses alter the shape of the face and adding resonance to the voice